Biofabrication

International Society for Biofabrication

PURPOSE-LED PUBLISHING™

PAPER • OPEN ACCESS

Leveraging transfer learning for efficient bioprinting

To cite this article: F Bracco *et al* 2025 *Biofabrication* **17** 035022

View the article online for updates and enhancements.

You may also like

- Osteochondral organoid biofabrication: construction strategies, applications and perspectives
Liwei Fu, Jiang Wu, Zhichao Zhang et al.

- Bioprinted *in vitro* tissue models: an emerging platform for developing therapeutic interventions and disease modelling
Nandana Bhardwaj, Souradeep Dey, Bibrita Bhar et al.

- Advanced bioprinting strategies for fabrication of biomimetic tissues and organs
Wei Long Ng, Cian Vyas, Boyang Huang et al.

This content was downloaded from IP address 95.97.24.146 on 15/09/2025 at 12:54

Biofabrication

CrossMark

OPEN ACCESS

RECEIVED
13 April 2025

ACCEPTED FOR PUBLICATION
19 June 2025

PUBLISHED
26 June 2025

Original Content from this work may be used under the terms of the Creative Commons Attribution 4.0 licence.

Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.

PAPER

Leveraging transfer learning for efficient bioprinting

F Bracco[1], G Zanderigo[1], K Paynabar[2] and B M Colosimo[1,*]

[1] Politecnico di Milano, Department of Mechanical Engineering, Via La Masa 1, 20156 Milan, Italy
[2] H. Milton Stewart School of Industrial and Systems Engineering, The Georgia Institute of Technology, Atlanta, GA, United States of America
[*] Author to whom any correspondence should be addressed.

E-mail: biancamaria.colosimo@polimi.it

Keywords: transfer learning, 3D bioprinting, extrusion bioprinting, machine learning, printability, in-situ monitoring

Abstract

Bioprinting is a promising family of processes combining 3D printing with life sciences, offering the potential to significantly advance various applications. Despite numerous research efforts aimed at enhancing process modeling, optimizing capabilities, and exploring new conditions, there remains a critical need to enhance process efficiency. Experimental data are paramount for improving models. Nevertheless, it is practically unfeasible to explore a multitude of conditions (e.g. different material formulations, process parameters, machines, setups), especially given the experimental constraints of budget and time. Leveraged by in-situ bioprinting monitoring, this paper explores a set of transfer learning (TL) methods designed for resource-efficient bioprinting modeling, aiming to merge established knowledge with new experimental conditions. TL encompasses machine learning strategies focused on transferring knowledge across distinct, yet similar, domains. TL is applied to an extrusion-based bioprinting case study for printability response modeling. The knowledge acquired from a model trained on one material (the source) is transferred to a new material (the target), under conditions of limited experimental data availability. Eventually, the accuracy of the transferred model is assessed and compared against a reference no-transfer scenario, which is developed from scratch following conventional practices. Furthermore, giving high importance to the experimental effort reduction, a sensitivity analysis altering the number of experimental training points is performed to assess performances and limitations of the method. This method demonstrates the feasibility of knowledge transfer in bioprinting as a catalyst for more sophisticated applications across diverse printing conditions, materials, and technologies to advancing this technology towards achieving its full potential.

1. Introduction

3D bioprinting (3DBP) is an emerging biofabrication technology with diverse applications, covering fields such as life sciences, tissue engineering, drug delivery, food bioprinting and space applications [19, 37, 39, 41, 58]. Various techniques are employed in 3DBP typically categorized based on the nature of the process: physical deposition of nozzles (including extrusion and inkjet) and optical based (such as laser-assisted, vat photopolymerization and two-photon polymerization) [58]. Among these, extrusion-based stands out as the most prevalent, extensively studied in literature [21], and commercially available [58]. Materials suitable for biofabrication, including synthetic, natural, or combinations of both, require a formulation that consistently strikes a delicate balance between printability and cell compatibility [18, 42, 59]. These materials, referred to as bioinks, constitute alone a core interest of research within the field, due to the infinite array of options available [18, 24, 27, 28, 54]. Currently, several authors agree on the necessity of expanding the current limited library of biomaterials if we want to replicate the complexity in native tissues in terms of multiscale organization and multimaterial composition [9]. However, the growing number of available materials and technologies significantly increases the workload for process or material modeling and optimization [66]. Given that the geometric and structural fidelity of the final construct is crucial to mimic the complexity of tissues and allows full biological functionality to be reached, the

© 2025 The Author(s). Published by IOP Publishing Ltd

evaluation of printability has gained importance in the field of bioprinting [18, 20, 42, 59]. In recent years, several techniques and methods have been implemented to increase the process qualification, enlarge the printability window, and thus push 3DBP towards advanced applications. The trial-and-error approach, typical of early bioprinting, has been overcome by systematic methodologies for the assessment and prediction of printability [31, 53]. The printing results are always influenced by several factors, sometimes uncontrollable and difficult to measure, involving the process (cross-linking strategy, process parameters), the bioink (rheological properties, physical properties, formulation), the design of the structure, and external conditions [18]. Various types of predictive models have been used in the literature for diverse tasks. The purposes include: sample design [53], identification of a feasible biofabrication window [7, 48], material selection and optimal formulation [17, 33, 65], cross-linking optimization [33], printing parameters optimization [3, 8–10, 23, 33, 53, 74], cell viability prediction and maximization [22, 38, 65], in-situ quality monitoring [10, 25, 35, 40]. These models can be either physics-based [7, 52, 67, 68], incorporating only the material properties, fully empirical [4, 5, 26, 70, 74], or mixed models [70]. Furthermore, the assistance of Design of Experiment (DoE) methods has promoted a more systematic approach for process modeling and material development [2, 48, 71, 74].

Finally, artificial intelligence (AI)-driven approaches are emerging as promising tools for optimizing bioprinting processes and enabling future advancements in the field [77]. Bone *et al* [11] demonstrated the optimization of bioprinting using small datasets, effectively addressing critical bottlenecks in experimental scalability and precision within bioprinting systems. While Jin *et al* [29] leveraged fine-tuning of a pretrained model to detect anomalies from transparent bioink images, thereby overcoming the limitations posed by the scarcity of large training datasets.

In light of this, three key considerations arise. (i) Physics-based models serve as a foundational approach. However, experimental data are typically required to calibrate these models and account for additional behaviors that cannot be captured through theoretical modeling alone. (ii) The extensive experimental effort needed to address the vast array of process conditions, materials, machines, and procedures presents significant challenges in developing general models across diverse scenarios. (iii) The access to extensive datasets is limited by constraints, especially cost of experiments or time.

In this paper we exploit machine learning (ML) aiming at transferring knowledge from one domain to another via transfer learning (TL). In the manufacturing context, domains are referred to context where the design, material or processing machines and conditions are varying (i.e. using a different product design, a different class of materials, a different machine or process strategies. In the general TL framework, established knowledge in a domain, referred to as the 'Source', is re-used and transferred to a 'Target', which is a similar but not identical domain [43]. TL is a set of ML techniques that have been demonstrated crucial when there is a limited supply of target training data, useful in several contexts to solve both regression or classification problems [43, 72, 73, 79]. The advantage of TL over traditional ML methods lies in its assumptions: TL allows for the use of data from different distributions (and thus different domains) to train a model, whereas traditional ML methods require that training and testing data come from the same distribution [43].

The figure 1 shows a schematic overview of a classical TL scenario in a manufacturing context. In this area, different works have been proposed for efficient process modeling, optimization, or qualification under varied conditions. According to the classification made by Zhang *et al* [76], TL has been used in additive manufacturing (AM) context for different purposes: leveraging AM-related datasets, both source and target are collected from experimental activity or from literature; AM-unrelated datasets, in which a pre-trained general model is fine-tuned using target data; multi-task learning, a co-learning scenario from multiple targets. Examining the first application in detail, TL has been used to adapt or migrate models for different materials [1, 16, 45, 63, 64, 75], process settings [14, 34, 44, 60, 75], design or shape [12, 13, 15, 50, 55, 56], machines [1, 16, 30, 51], processes [36, 57, 78].

The aim of the current work is to investigate the potential advantages of TL in the framework of bioprinting, specifically addressing one of the most common challenges: process optimization when the bioink needs to be changed. Given the extreme variety of bioinks that can be designed and the high sensitivity of printability to material changes, we believe that this framework can have a significant impact on the bioprinting community. Our goal is to evaluate and compare the performance of TL methods against the classical approaches commonly used, quantify the prediction accuracy of both, and draw general conclusions on the practical usefulness of TL for resource-efficient process optimization in 3DBP. In particular, the current paper focuses on the most common 3DBP technology, namely extrusion, where a printability response model realized for a material (M1), namely the Source, is transferred to another one (M2), the Target. Our aim is to show that by transferring the pre-existing knowledge acquired on the source material M1, a significant reduction of the experimental effort on the target material M2 can be achieved.

The main contributions of this paper are: (i) demonstrating the significant benefits of applying knowledge transfer to reduce experimental effort when exploring a new bioprinting conditions, namely

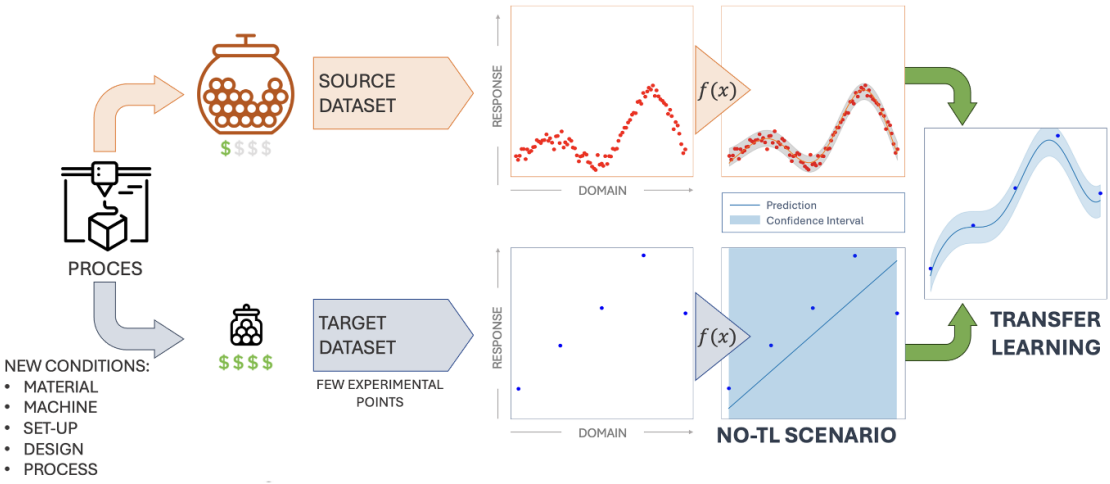

Figure 1. A general overview of the Transfer Learning scenario in a manufacturing context. The learning performance of a target task (in this case exemplified by a single-input response model) is enhanced by incorporating knowledge from a similar but not identical source domain. This latter is considered as a cheap source of knowledge since it has been already established, while the target is represented by a dataset collected under new conditions. In this way, thanks to the transferred knowledge, the resulting target model, even when trained with limited experimental observations, features better prediction performance than a model trained from scratch, using only the target dataset.

exploring printability of a differente bioink; (ii) evaluating the effectiveness of TL through a sensitivity analysis, highlighting its impact relative to the experimental effort required in the target domain; and (iii) showcasing its practical utility with a real-world example, which illustrates the superior accuracy and robustness of the proposed method compared to traditional approaches used for process modeling.

The paper is organized as follows: section 2 presents the methodology employed, describing firstly the problem setting and encompassing all the phases of the current work from the experimental campaign in sections 2.2 and 2.3 and a description of the proposed knowledge transfer framework in 2.5. Results are shown and discussed in section 3, starting from a case study that serves as a simple example to illustrate the practical implications of the method and, then, conducting a deep performance analysis of the proposed approach. Final remarks and further considerations in the concluding section 4.

2. Materials and methods

2.1. Problem definition

The motivation for this work is to determine the optimal process parameters to enhance the printability of a biomaterial, denoted as M2 (the target material), which has never been printed before. However, prior knowledge exists regarding the effect of process parameters on the printability of a different material, M1 (referred to as the source material). The main objective of TL is to investigate how the pre-existing knowledge of the source material (M1) can be used to develop a printability model for the new target material (M2) in a resource-efficient manner, that is, with reduced experimental effort.

Following established practices, two process parameters have been considered relevant for the bioprinting assessment: the extrusion pressure (P) and speed (V). The quality response is represented by an indicator of shape fidelity, generically referred to as the printability index, which can be directly evaluated from in-situ images of the printed construct and is defined as:

$$PI = \frac{1}{4}\sum_{i=1}^{4}\frac{A_i}{A_{\mathrm{Ref}}} \tag{1}$$

figure 2 shows the main steps of the proposed procedure.

A grid-based full-factorial (DoE) was initially implemented for the source material M1, considering six levels for both parameters P and V, with three replicates per condition. This results in an experimental plan with $n = 6 \times 6 \times 3 = 108$ trials to estimate the printability map for M1 (see figure 3).

For the target material M2, the same experimental plan of $n = 108$ was conducted; however, the data set was split into a training and a validation sets. Specifically, a training dataset of $m = 10$ data points was used to estimate the printability index for M2, both with and without TL. This training set followed a 3×3 factorial layout, including a replicated centroid, as illustrated in figure 3.

The remaining $n - m = 108 - 10 = 98$ data points comprised the validation set, used to assess the performance of the models trained on the reduced

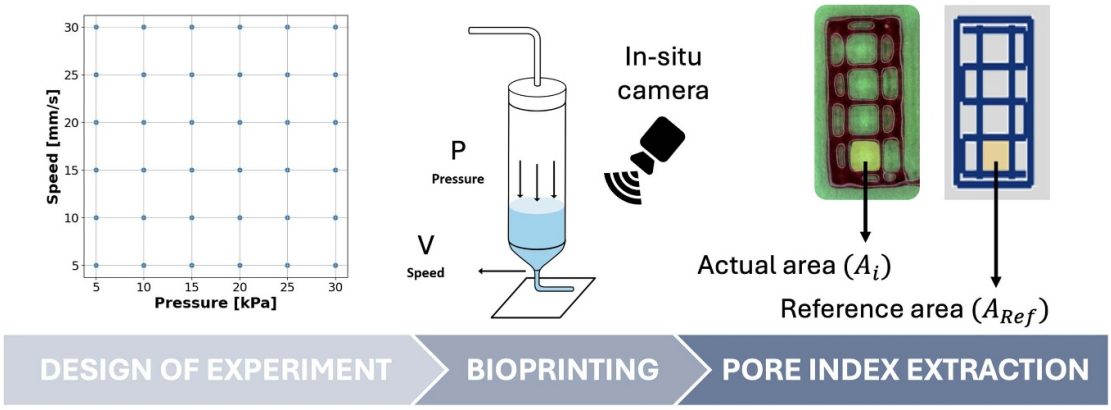

Figure 2. Schematic overview of the experimental workflow: from left to right the bi-dimensional domain and parameters of the design of the experiment, the extrusion-based bioprinting process monitored utilizing an in-situ camera, and the PI index extraction. This was conducted measuring the average void area in each construct and comparing it with the original design.

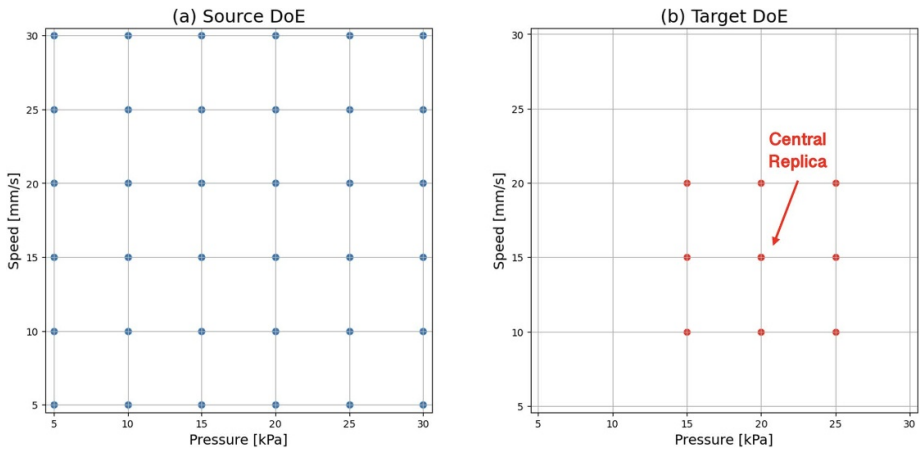

Figure 3. The experimental datasets for the (a) source and the (b) target. (a) Source domain: $n = 108$ experiments, given by $6 \times 6 = 36$ experimental conditions replicated 3 times. (b) Target domain: $m = 10$ experiments given by $3 \times 3 = 9$ experimental conditions with 2 replicates at the central point.

m data, with and without TL. Notably, the validation set followed the same full factorial design of $n = 108$, allowing a fair performance comparison.

As summarized in figure 4, the training and validation strategy prioritized resource-efficient modeling, with a training set significantly smaller than the validation set, contrary to the standard practice in ML where large data sets are usually considered for training. Finally, a sensitivity analysis was performed to explore the benefits of TL as the size of the training dataset varied. The details and results of this analysis are eventually presented in section 3.1.

A summary of notation and details of the experiments considered in this paper are shown in table 1, where matrices are denoted by bold uppercase letters, and vectors by bold lowercase letters.

2.2. Materials and setup

Two bioinks with a common component were used, namely Alginate 6% (referred to as the source material with the corresponding predictive model, M1) and Alginate-Gelatine 6%–4% (referred to as the target material with the corresponding predictive model, M2). The materials were chosen because they are commonly used in bioprinting and exemplify a typical knowledge transfer problem, where the target domain is similar, but not identical, to the source domain. Both bioinks were based on the same polymer backbone, namely alginate, with gelatin added in the second formulation. The latter was identified as the target because it is of significant interest, within a conventional research framework, to efficiently explore a new formulation by leveraging an established base polymer. To prepare the bioinks, two stock solutions were created. An 8% ($\mathrm{w\,v^{-1}}$) Sodium Alginate stock solution was formed by dissolving Sodium Alginate (Sigma Aldrich) in sterile PBS, which was then continuously stirred at 50 °C for 4 h using a heated magnetic stirrer. Additionally, a 16% ($\mathrm{w\,v^{-1}}$) Gelatin stock solution was made by dissolving Gelatin (Sigma Aldrich) in sterile PBS and stirring the solution at 30 °C for 4 h with a

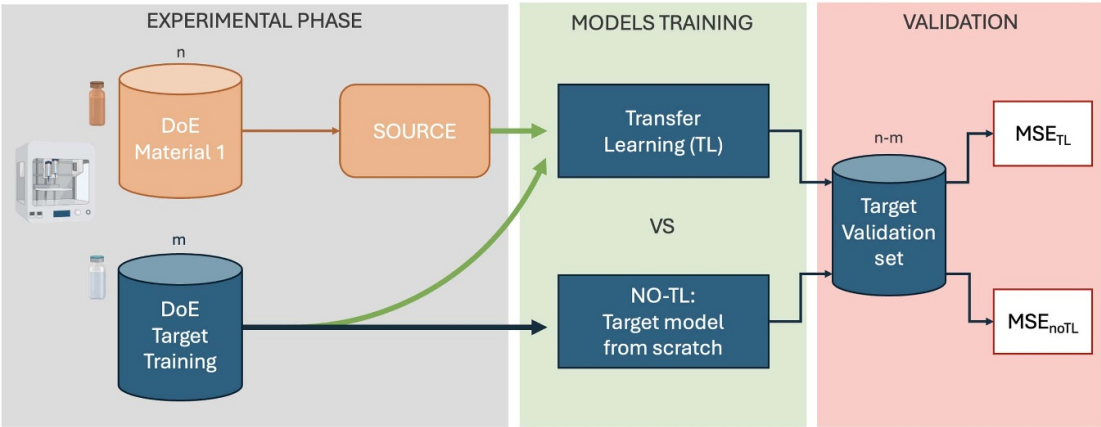

Figure 4. General procedure framework to apply and assess transfer learning in the work. TL model was trained leveraging the source knowledge with the target training dataset. Only this latter, instead, were used to train the reference approach model from scratch. The validation was made comparing the MSE of the two scenario using a dedicated validation dataset.

Table 1. Description of notations used in the study.

Notation	Description
M1	Alginate 6% (Source material)
M2	Alginate-Gelatine 6–4% (Target material)
P	Extrusion pressure. Process parameter explored
V	Scanning Speed. Process parameter explored
PI	Pore Index. Quality indicator observed in experiments
X	Input domain. Each instance of the domain represents a pair of P and V values
y	Response
D	Dataset. It has both input and output data as columns: $D = [X\|y]$
$f()$	Printability response surface model
Starred X_*, y_*	Predictions
Subscripts $_t$ and $_s$	Quantity or feature associated respectively with Source or Target
n and m	Dimensions of Source and Target datasets

heated magnetic stirrer, following the standard protocol provided by the supplier. These solutions were refrigerated overnight for complete hydration. The first bioink was produced by diluting the 8% Alginate stock solution and PBS in a 3:1 ratio, while for the second bioink the 8% Alginate stock solution was dissolved with Gelatine stock in 3:1 ratio. Moreover, working solutions contained 10% ($w\,v^{-1}$) Amaranth solution to give the inks a vivid red color and help the segmentation phase. The bioprinting platoform used was the BIO X™ (Cellink®) equipped with a pneumatic extruder with temperature control. The printer was loaded with 3 ml cartridges (Cellink®) featuring a 0.410 mm diameter conical nozzle (Cellink®). In-situ images were captured using the HD Camera Tool (Cellink®). An external ring light was used during image acquisition to enhance image quality. For more detailed information on experimental setup refer to [74].

2.3. Experimental data collection

A grid-based full-factorial DoE was conducted exploring the two parameters P and V at 6 levels each: $V \in [5, 10, 15, 20, 25, 30]\,\mathrm{mm\,s^{-1}}$ and $P \in [5, 10, 15, 20, 25, 30]\,\mathrm{kPa}$, as shown in figure 2. Each combination of parameters was randomly replicated 3 times, for a total of 108 experimental observations. The domains, limits and steps were chosen according to the preliminary experience and considerations on the process stability. Moreover, domains were chosen in order to observe complete over and under-extrusion for both the materials. The bioprinted construct was a mono-layer grid-like structure of dimensions $15 \times 16 \times 0.4\,\mathrm{mm}$. The HD in-situ camera was used to capture the bioprinted construct and images were analyzed offline to quantify the PI . This index was defined as the average ratio of the central 4 voids in the grid within each construct with respect to the reference area, as in 1.

The image analysis was conducted using the Region growing algorithm (MATLAB R2022a) [32] for void area segmentation. The optimal value of the PI is 1, while output value ranges by construction between 0 (extreme over-extrusion) to 3 (complete under-extrusion). The upper limit value comes from the dimension of the maximum area in which contours of voids are searched by the algorithm in the images. After the analysis, the datasets were arranged

in structured tables with 108 rows and 3 columns, two regressors (P and V) and one output y, representing the measured PI values. Figure 2 shows a representative overview of the workflow: from left to right the scheme of domain explored, bioprinting process phase, and the index extraction.

The training set for the second target material M2 was designed considering a typical procedure, which involves an initial rough identification of a central point, which is arbitrarily chosen close to the optimal printability line identified with the source model, figure 3(a). Once the central point was defined, experiments were planned in its vicinity. In this case study, we center the experiment around the point $P = 20\,\text{kPa}$ and $V = 15\,\text{mm}\,\text{s}^{-1}$ with a 2×2 plan in the surrounding region, including a central replica, as shown in figure 3. The $m = 10$ observations were used to train an OLS model from scratch (no-TL scenario) and a TL model, following the approach presented in section 2.5. The remaining set of $n - m = 98$ experimental runs were eventually used as validation data set. Details are provided in the appendix, table A1.

2.4. Gaussian processes for response modeling

The experimental observation were used to train models for learning the response surfaces, and make predictions using surrogate models based on Gaussian process (GP). GPs are defined as a collection of random variables, any finite of which have a joint Gaussian distribution [49]. GP were chosen for their flexibility in describing observed patterns and for representing a general approach that can be easily extended with more complex response functions. Furthermore, due to their probabilistic nature, it is possible to readily compute confidence intervals for predictions at any new points. A GP is fully characterized by its mean $m(\mathbf{x})$ and covariance $k(\mathbf{x}, \mathbf{x'})$ functions and it is denoted as $f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x'}))$. The covariance function specifies the covariance between a pair of random variables. In this work, the squared exponential kernel was chosen, which defines the covariance based on the distance between the respective random variables. This choice represents common practice and provides maximum flexibility and smoothness in the solution, as well as ease in computing the inverse of the covariance matrix. Squared exponential introduces two parameters in the models for the distance length scale ℓ and variance σ_ℓ^2. Additionally, a Gaussian noise term with variance σ_y^2 is added to account for the experimental variability. Concerning the mean structure, we utilized zero mean prior belief to allow for maximum flexibility. Parameters of models were computed maximizing the log-marginal-likelihood employing the built-in *scikit-learn* library—ML in Python [47]. Given the posterior and the estimate of parameters $\hat{\boldsymbol{\theta}}$, GP surrogate model allows for predictions $\mathbf{y}_*$ at each test point $\mathbf{X}_*$. The starred letters indicate predictions at the new test points. More formally, the new test points are added to the joint multivariate Gaussian distribution with training data, see 2

$$\begin{bmatrix} \mathbf{y} \\ \mathbf{y}_* \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} K(X,X) + \sigma_n^2 I & K(X,X_*) \\ K(X_*,X) & K(X_*,X_*) \end{bmatrix}\right). \quad (2)$$

Notice that the predictions at new points based on GP are not computed from a closed predictive model in a parametric form, but obtained as collection of random variables with estimated joint distribution. For the remainder of the paper, the trained GP surrogates will be denoted as predictive models $f(X_*)$ or simply $f(X)$, although this notation is not entirely correct. The correct representation should employ conditional probability notation of the estimated instances: $f_* \mid \mathbf{X}_*, \mathbf{X}, \mathbf{y}, \hat{\boldsymbol{\theta}}$. In figure 7, the predictions of GP surrogate models are shown, bounded within the entire domain explored in DoE for both M1 and M2. Both models were trained using the respective 108-sized entire experimental dataset. The M2 model was trained just to compare the materials' behavior by analyzing their response surfaces. Comparison between the two models revealed that M2 exhibits a smoother behavior compared to M1. In M1, a steeper gradient was observed in the central band of the domain, with a wider area of under-extrusion for the lowest pressure values. The difference in viscosity between the two materials results in variations in the optimal printability area across a PI of 1. This area occurs for lower pressure values and higher speeds when transitioning from M1 to M2.

2.5. Transfer learning

Knowledge transfer was applied assuming the n-sized source and the m-sized target training subsets to be available. Prediction performances with and without TL were compared assuming $n - m$ data available as validation set. Among the various TL options [43], the instance-based (IB) solution was chosen as a reference for this study, given its well-stabilized popularity when the source and target domains have a similar space. With IB, the knowledge that comes from each single source data point, known as instance, is valued in the experiment using an importance or weighting strategy [43]. Therefore, the method enables the migration of knowledge from the source to the target domains by selectively utilizing only the most relevant source of information, where relevance is given by resemblance to the target, with the intent of minimizing the impact of data points of the source domain that are very different from data observed in the target one. As several different algorithms can be used for the IB-TL, in this study we developed a custom algorithm, starting from the existing literature [6, 46, 61, 62, 69]. The

algorithm integrates the source and target datasets using a GP weighted regression strategy, solved iteratively. Briefly, the algorithm starts with an initial weighted regression model trained on both the source and the target domains, using a simple binary guess for the initial weights of the source data. The model parameters and weights are then iteratively updated to minimize the error, until convergence is reached. A detailed description of the algorithm and references used can be found in the supplemental material appendix.

The proposed developed TL solution was, eventually, compared to the no-transfer scenario. The no-TL predictive model was trained from scratch using only the target data, without considering any pre-existing knowledge gained thanks to the source domain. This no-transfer scenario is implemented as reference to represent the current practice. Specifically, the no-transfer scenario considers two types of model fitting for the sampled data, namely GP and linear ordinary least square (OLS) up to the second order.

3. Results

Results of our investigation are eventually discussed to explore the feasibility and efficiency of TL in a real-world application scenario. With reference to the bioprinting experiment described in section 2.3, $n = 108$ experimental trials for material M1 (source domain) and $m = 10$ data points involving material M2 (target domain), were considered to train the TL as well as the no-TL models. Subsequently, an additional dataset of $n - m = 98$ data points was used to validate the prediction efficiency with and without TL.

In order to show the effectiveness of the proposed TL approach, predictions of the printability index in the target domain were compared with the $n - m$ validation data to compute the mean squared error (MSE), given by:

$$\mathrm{MSE} = \frac{1}{n-m}\sum_{i=1}^{n-m}\left(y_i - y_{\mathrm{validation}_i}\right)^2. \qquad (3)$$

Table 2 shows the resulting MSE values.

The 3D response surfaces of the printability index estimated with and without TL are shown in figure 5, together with the training and validation data sets observed in the target domain. The advantage of TL can be easily observed by comparing the plots, as the response surface estimated without TL can be quite far from the real observations in some locations of the printability map. Indeed, the TL model better fits the real dataset due to the incorporated knowledge from the source domain. This is particularly clear in regions of the domain that are far from the small-scale experiment conducted in the target domain using the m training data only. In this case, TL enables extrapolation with greater confidence.

Table 2. Prediction performances of the no-TL and TL models. MSE (equation (3)). The lowest values are highlighted in bold.

	no-TL	TL
MSE	1.1075	**0.1979**

Additional perspectives can be also highlighted considering the prediction intervals obtained for the printability index with and without TL. To enhance visualization, a detail of these prediction interval is shown at a fixed level of pressure $P = 20\,\mathrm{kPa}$ in figure 6. At this level of pressure, the best value of speed should be around $25\,\mathrm{mm\,s^{-1}}$, as this is the location where the printability index is equal to 1. With no-TL (6(b)) the fitted line (continuous line—light Grey) is very far from the real one (dotted line), especially in the region close to the optimal value of speed ($25\,\mathrm{mm\,s^{-1}}$). On the other side, using the TL approach (6(b)), the fitted line (continuous line—light Grey) is quite close to the actual one (dotted line). The prediction intervals also includes almost all the line at PI = 1, i.e. any Scanning Speed within the domain would be a possible candidate.

In contrast, only 10 experimental points are sufficient when knowledge transfer is applied to obtain a quite accurate model, which highlights that the prediction interval intercepts the PI = 1 line quite close to the optimal Scanning Speed that is indeed 25 $\mathrm{mm\,s^{-1}}$.

To better investigate the effectiveness of the TL solution, the contour map of the printability index in the source (M1) and target (M2) domains fitted considering all the available datasets of $n = 108$ experiments in each domain, are shown in figure 7. As can be noted, the source and target domains have different shapes of the iso-curve of printability, even if located in similar zones of the $P - V$ printability space. What TL does is basically transferring the previous knowledge of the printability map and locally readapting this knowledge in target domain, even when a reduced amount of information is available.

3.1. Sensitivity analysis

In order to better compare the performances of the TL approach and deeply investigate its effectiveness, a sensitivity analysis was eventually conducted. In this case, we again decided to split the dataset in the target domain into a m-sized training and $n - m$-sized validation sets. Here we decided to vary the training data set size $m \in \mathbf{M} = [4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 18]$ and for each value of m, the analysis was repeated 100 times by randomly selecting the training data set. In each of this replicate, for each scenario, the MSE was computed according to equation (3), using the $n - m$ validation data remained . The resulting prediction accuracy of the TL models versus the noTL

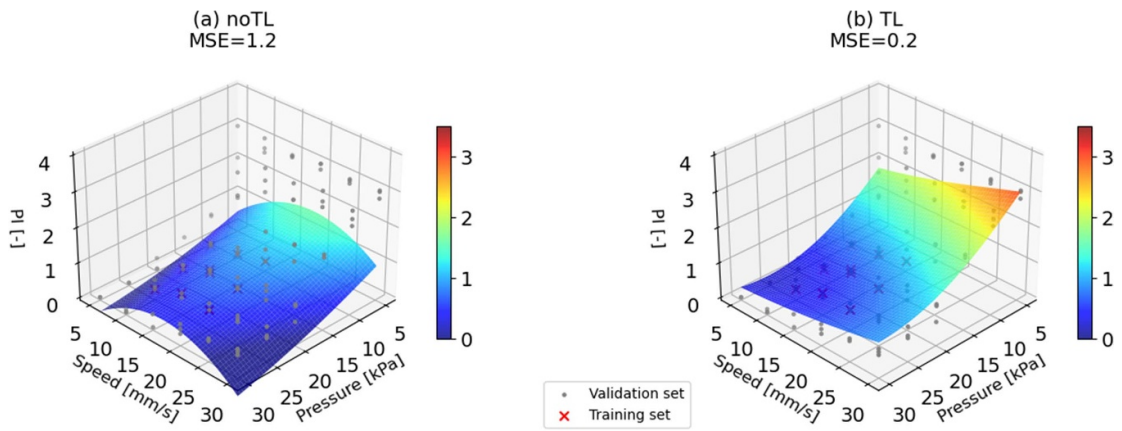

Figure 5. 3D surfaces of the resulting models, respectively (a) no-TL and (b) TL. The target experimental dataset, split in target testing dataset (gray) and target training (red), are shown in the domain.

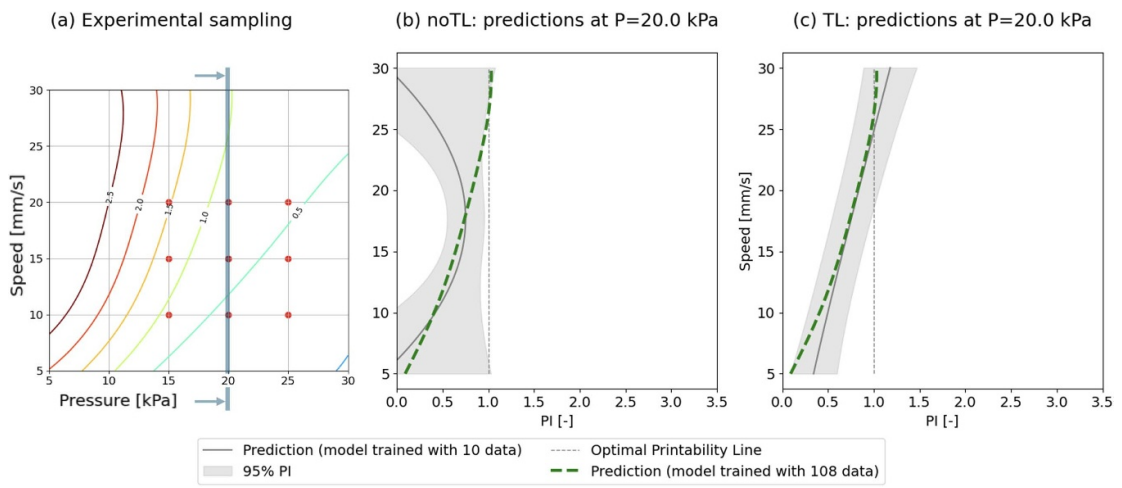

Figure 6. (a) The 10-points experimental design is superimposed on the real isomaps of printability index (obtained using the whole data sets on material M2). Predicted mean (dashed line) and 95% prediction interval (gray band) for (b) no-TL and (c) TL models in a section of the printability domain (at fixed $P = 20$ kPa).

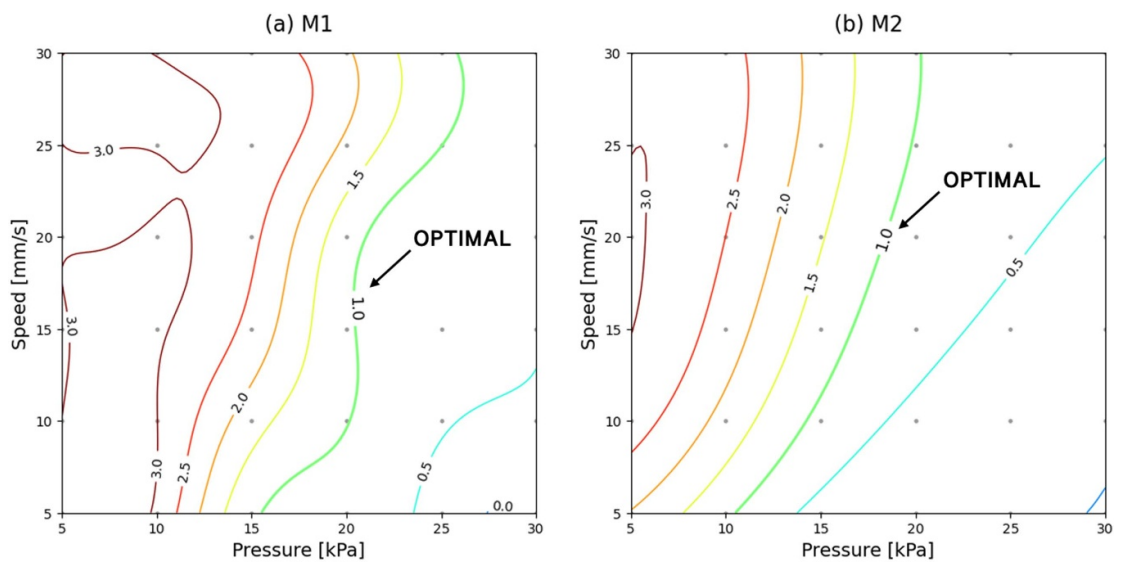

Figure 7. Iso-curve of printability index for the (a) source M1, and (b) target—M2 domains estimated using all the available set of $n = 108$ data. Curves are here shown just to observe the difference between the source and target domains.

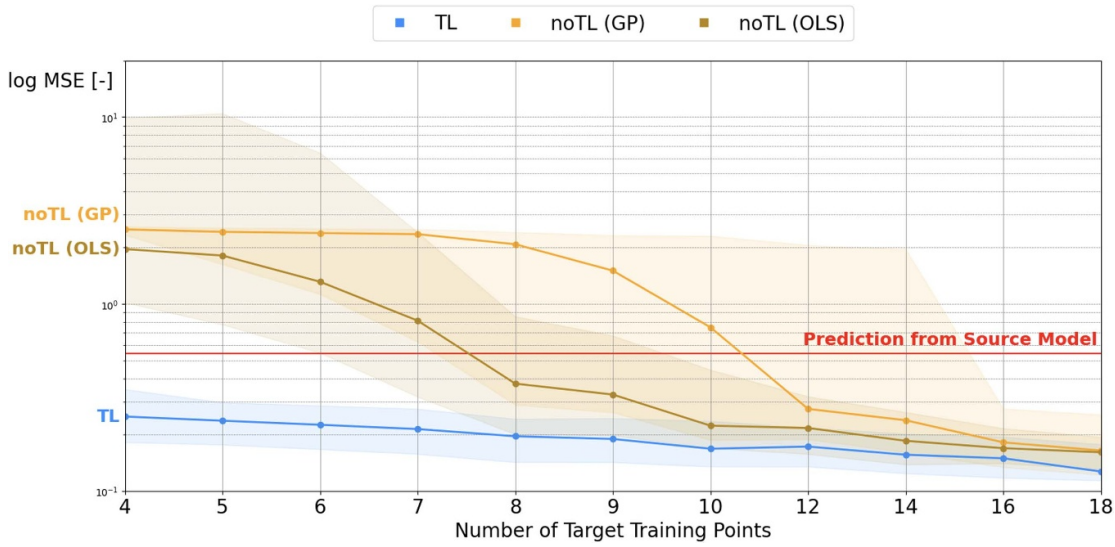

Figure 8. Prediction performance as number of target training points varies of the TL model against no-Transfer scenario—GP and OLS models. The central line represents the median over 100 replicas and the shaded regions represent the two central quartiles (Q25-Q75 percentiles). On the vertical axis, the MSE computed on the target testing dataset on a logarithmic scale. The red line represents the experimental prediction error made using directly the source model.

approaches is shown in figure 8 on a logarithmic scale, as the number of target points varies. The shaded areas represent the data within the 25th and 75th percentiles, across 100 repetitions. The horizontal red line represents the direct prediction from the source model, which should be appropriate under the assumption of no differences between the target and the source. Table A2 shows the same in details. For the problem approached in this paper, the TL model performed better predictions than using the source model directly for any number of target points m. In such a scenario, knowledge transfer always results in a positive gain in knowledge, even if this can be very problem-specific. The most interesting aspect of the plot is the performance with the lowest number of target points. Even with a very small sample size in the given problem, the TL approach offers a valuable and powerful method to build a reliable model with controlled prediction error. Additionally, it is evident that for fewer than $m = 10$ target points sampled, the GP trained from scratch struggles to produce a reliable predictive model. Therefore, in such cases, it is more appropriate to use the OLS model as a comparison reference, as implemented in the case study.

Finally, besides the bias, the variance is another important behavior to be taken into account for a robust predictive model. TL exhibited much narrower prediction intervals compared to the no-TL. This is the outcome of a proper training of TL models which properly incorporates more data, and thus more knowledge, from the source domain.

4. Conclusion

The presented study introduces a TL framework tailored for resource-efficient modeling of bioprinting processes. While the achievements made are modest, they address critical challenges outlined in the literature, particularly those related to improving process modeling, optimizing capabilities, and exploring new conditions in scenarios with limited data availability. Key data-related obstacles in bioprinting, already highlighted in the literature, can be summarized as follows.

- Data size: Intensive experimental campaigns or procedures are required for data collection to train models, especially deep learning models which necessitate large datasets. This results in significant material waste, whether for training predictive models or for the trial-and-error approach common in bioprinting.
- Data cost: Both high costs and significant time are required for experimentation, particularly with biological cell-laden materials, niche technologies, and under extreme printing conditions.
- Data quality and availability: Integrating data from the literature is hindered by its scarcity (less than 30% of experimental papers provide data [9]) and inconsistency (datasets collected from diverse bioprinters, processes, platforms, and operational protocols).

Classical ML represents a significant turning point to address bioprinting process qualification and optimization. However, in scenarios with limited data availability, traditional ML methods struggle to create trustable models. Furthermore, the incorporation of new data to train models, whether from literature or different experiments, can be detrimental as traditional ML assumptions dictate that the data must come from the same probability distributions, both marginal and conditional. In contrast, TL relaxes this constraint and enables knowledge transfer through appropriate methods. By addressing data-related challenges through a real case study, we demonstrated the efficacy of knowledge transfer in constructing accurate models with limited experimental data, thereby reducing costs and time associated with experimentation. Our analysis also incorporated the sensitivity of the method to varying number of experimental training points, emphasizing the importance of reducing experimental effort and highlighting both its strengths and limitations. Importantly, our findings highlight the significance of TL, particularly in scenarios with sparse data across the domain, as conventional approaches often fall short due to insufficient sampling or reliance on data from different domains. Moreover, the explanatory example clearly demonstrated the practical implications of knowledge transfer in a real bioprinting scenario, where the proposed TL method outperformed the conventional approach in process modeling and prediction. It provided improved confidence in the accuracy and enhanced capacity for extrapolation beyond the experimental domain. Despite the results, further demonstrations of the knowledge transfer approach must be achieved by exploring different materials and additional case studies.

These insights hold important implications for the field of bioprinting, offering a pathway towards more sophisticated applications across diverse conditions, more complex models, new materials, and various technologies. Moving forward, the integration of TL methodologies can help the learning of new tasks and revolutionize bioprinting research by unlocking new avenues for innovation, addressing the challenges inherent in replicating biological complexity.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://doi.org/10.1016/j.bprint.2023.1000320.

Acknowledgments

This study was carried out within: 'HORIZON-CL4-2021- DIGITAL-EMERGING-01 Project BioProS—Biointelligent Production Sensor to Measure Viral Activity' (Grant Agreement No. 101070120), 2022–2026. The Authors acknowledge Davide Moscatelli and his support in the 3D Cell Lab.

Appendix. Supplemental material

A.1. Transfer learning method: instance-based

In the IB TL approach, the knowledge coming from each single source data point, referred to as instances, is utilized in the model by leveraging the importance of each separately, using an importance or weighting strategy [43]. In this way, the approach allows to gain only useful information from the source instances that are most similar to a target and almost ignore the others. One of the most commonly used IB-TL algorithm is the so-called the Two-stage TrAdaBoost.R2 algorithm (2sTAB.R2) [46]. Nevertheless, it has been used with poor result for similar problems [75]. We implemented and used a modified version of the IB approach, leading to a more flexible method, more accurate and compatible with the GP modeling. The method shares the same rational with the previous work presented in [46], which is based on an iterative weighting strategy. The instance importance information is combined in the GP modeling framework within the covariance structure, trough a *noisy biased kernel design* strategy [6, 62, 69]. Differently than [62, 69], where importance is assigned to the entire source dataset as a unique value to leverage it with the target, our proposed approach assigns an importance weight to each instance in the source dataset, providing maximum flexibility to the method. The idea of single instance weighted GP was used outside of TL context to get rid of outliers in GP modeling framework by Sheng *et al* [61].

The details of the IB algorithm proposed are summarized in the algorithm 1 and detailed as follows.

Firstly, dataset of source and target must be pooled together in an unique dataset: $\mathbf{D} = \{\mathbf{D}_t, \mathbf{D}_s\}$. The importance of instance was leveraged by means of a vector $\mathbf{d}$, representing scaled metric distance between instances and the prediction of the TL model at the same location. The vector was initialized to give maximum importance to target at first, and, then, updated after each iteration. At the convergence, an estimated of TL model and distance vector are obtained. The latter was the absolute prediction error of the obtained model on actual dataset. The GP model was defined and updated using the distance information in the covariance structure, as defined in A2. The key point of the method is the effect on the kernel of the distance vector and the α parameter, acting as an additional noise variance associated to each instance in two different ways. The distance vectors acts individually on each instance, reducing its effect on resulting model for increased value, while the α has a global effect on the overall source dataset, leading to completely neglecting its

Algorithm 1. Instance-based Transfer Learning Algorithm.

1: **Input:** Source dataset $\mathbf{D_s} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$, Target dataset $\mathbf{D_t} = \{(\mathbf{x}_j, y_j)\}_{j=1}^{m}$, Initial distance vector $\mathbf{d}^{(0)}$ (equation (A1)), Scaling parameter α
2: **Output:** Final transferred model $\mathcal{M}_{\mathrm{TL}}$, Final distance vector $\mathbf{d}$
3: **Step 1:** Pool the source and target datasets together: $\mathbf{D} = \mathbf{D_t} \cup \mathbf{D_s}$
4: **Step 2:** Initialize the GP model using only the target dataset $\mathbf{D_t}$
5: **repeat**
6: **Step 3:** Use the current GP model to compute predictions $\mathbf{y}_*^{(t)}$ for the pooled dataset $\mathbf{D}$
7: **Step 4:** Update the distance vector $\mathbf{d}^{(t+1)}$ for source instances based on prediction error (equation (A3))
8: **Step 5:** Update the GP model using the new distance vector $\mathbf{d}^{(t+1)}$ (equation (A2))
9: **Step 6:** Check convergence:
10: **If** $\sum |\mathbf{d}^{(t+1)} - \mathbf{d}^{(t)}| < toll$ **or** $t \geqslant NMAX$
11: **Then** Stop iteration
12: **Else** Set $t = t + 1$
13: **until** convergence
14: **Step 7:** Return final transferred model $\mathcal{M}_{\mathrm{TL}}$ and distance vector $\mathbf{d}$

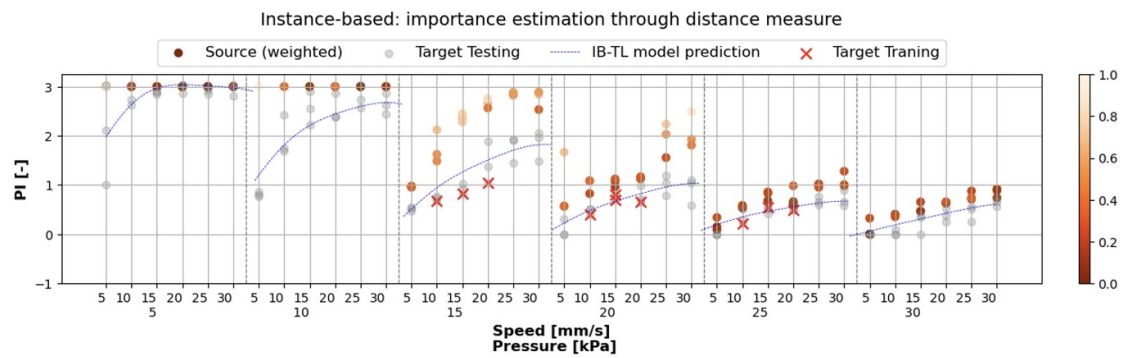

Figure A1. Visual depiction of Instance-based procedure on a value plot, which illustrate data in a 2D plot compressing the two domains in a single horizontal axis. The graph shows both the Source dataset (Material 1), the Target dataset, split in Testing (gray) and Training (red) points. The plot depicts the source data instances alongside their respective normalized distances from the model prediction. It is important to note that instances with greater distances (with lighter color) hold lesser significance in the training of the transferred model. Dashed line represents the prediction of the final IB-transferred model achieved. Additionally, it should be noted that the abscissa of the target points is not exact, but rather a projection derived from a linear interpolation.

contribution for $\alpha \to \infty$, or giving the same importance for source and target dataset for $\alpha = 0$. In the first case, the resulting model would be exactly the same model as trained with only target points, while the latter equivalent to a GP trained with both the datasets.

For this work, the scaling parameter was manually tuned in a pre-tuning phase and kept fixed $\alpha = 10$. Several values between 0.001 and 1000 were tested before ending up with chosen value. The initial distance vector $\mathbf{d}$ was initialized as in A1,

$$\mathbf{d}^{(0)} = \begin{cases} 0 & \text{if } i = 1, \ldots, m \\ 1 & \text{if } i = m+1, \ldots, m+n \end{cases} \tag{A1}$$

being m and n the length of target and source dataset respectively. The vector $\mathbf{d}^{(t)}$ was updated at any iteration t, with $t \in \mathbb{N}$, computing the absolute and normalized prediction error only on data corresponding to source, as in the A3. Finally, the GP model was updated as well with the current distance vector and so on until convergence

$$\begin{bmatrix} \mathbf{y} \\ \mathbf{y_*} \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} K(X,X) + \sigma_n^2 I + \alpha \cdot \mathrm{diag}\left(\mathbf{d}^{(t)}\right) & K(X,X_*) \\ K(X_*,X) & K(X_*,X_*) \end{bmatrix}\right) \tag{A2}$$

$$d_{i=m+1:m+n}^{(t+1)} = \frac{|y_i - y_{*i}^{(t)}|}{\max_{k=1}^{n+m}\left(|y_k - y_{*k}^{(t)}|\right)}. \tag{A3}$$

The iteration procedure was stopped when convergence of the distance vector was achieved or the maximum number of iterations reached. Convergence parameters chosen were: i) the tolerance $\mathrm{toll} = 1e-5$, computed as sum of absolute distances between distance vectors of the last two consecutive iterations; ii) the maximum number of iterations $t \leqslant \mathrm{NMAX} = 100$. Figure A1(a) shows a representative example of final distances computed on source instances to train a transferred model with $m = 6$ target points. It is crucial to highlight that instances with greater distances carry less importance in the model training process (indicated by dashed lines) as the iterations progress.

A.2. Target experimental dataset

Table A1. Experimental Target dataset, split into Training and Testing subsets.

Target subset	Pressure (kPa)	Speed (mm s^{-1})	PI [-]
Training	15	10	0.6781
	15	15	0.8314
	15	20	1.3772
	20	10	0.5189
	20	15	0.6986
	20	15	0.8043
	20	20	0.9953
	25	10	0.5055
	25	15	0.5042
	25	20	0.6811
Testing	5	5	1.0012
	5	5	2.1131
	5	5	3.0305
	5	10	2.7455
	5	10	2.6229
	5	10	3.0308
	5	15	2.8510
	5	15	2.9097
	5	15	2.8789
	5	20	2.8684
	5	20	3.0308
	5	20	3.0308
	5	25	2.9706
	5	25	2.9271
	5	25	2.8500
	5	30	2.8068
	5	30	2.9971
	5	30	3.0308
	10	5	0.8091
	10	5	0.7604
	10	5	0.8656
	10	10	1.6839
	10	10	1.7438
	10	10	2.4178
	10	15	2.2311
	10	15	2.5575
	10	15	2.9089
	10	20	2.3925
	10	20	2.3961
	10	20	2.8575
	10	25	2.5742
	10	25	2.7314
	10	25	3.0250
	10	30	2.4410
	10	30	2.6311
	10	30	2.8617
	15	5	0.4729
	15	5	0.5520
	15	5	0.5252
	15	10	0.7296
	15	10	0.7683

(Continued.)

Table A1. (Continued.)

Testing	15	15	0.9105
	15	15	1.0296
	15	20	1.0385
	15	20	1.8837
	15	25	1.4481
	15	25	1.9106
	15	25	1.9311
	15	30	1.9676
	15	30	2.0544
	15	30	1.4896
	20	5	0.0000
	20	5	0.0000
	20	5	0.3133
	20	10	0.5139
	20	10	0.3953
	20	15	0.7827
	20	20	0.7197
	20	20	0.6647
	20	25	1.0390
	20	25	1.1937
	20	25	0.7794
	20	30	1.0922
	20	30	1.0441
	20	30	0.5933
	25	5	0.0000
	25	5	0.0000
	25	5	0.0000
	25	10	0.2981
	25	10	0.2253
	25	15	0.5467
	25	15	0.4138
	25	20	0.4899
	25	20	0.4950
	25	25	0.7745
	25	25	0.5962
	25	25	0.6671
	25	30	0.5870
	25	30	0.6533
	25	30	0.9048
	30	5	0.0000
	30	5	0.0000
	30	5	0.0000
	30	10	0.0849
	30	10	0.0000
	30	10	0.0000
	30	15	0.3462
	30	15	0.3754
	30	15	0.0000
	30	20	0.5371
	30	20	0.2569
	30	20	0.3913
	30	25	0.5050
	30	25	0.5731
	30	25	0.2454
	30	30	0.6509
	30	30	0.7361
	30	30	0.5685

A.3. Accuracy results

The following table represents the data shown in figure 8.

Table A2. Median MSE (3) value on 100 replicas of the TL model and no-TL—GP and OLS—varying the number of target points sampled *m*. In bold, MSE values lower than the best reference model per row.

	Transfer Learning	no-Transfer	
m	IB-TL	GP	OLS
4	**0.2499**	2.5030	1.9636
5	**0.2374**	2.4283	1.8138
6	**0.2258**	2.3940	1.3138
7	**0.2142**	2.3597	0.8138
8	**0.1960**	2.0845	0.3747
9	**0.1896**	1.5060	0.3276
10	**0.1681**	0.7494	0.2233
12	**0.1727**	0.2745	0.2171
14	**0.1562**	0.2387	0.1853
16	**0.1495**	0.1818	0.1693
18	**0.1270**	0.1644	0.1609

References

[1] Aboutaleb A M, Bian L, Elwany A, Shamsaei N, Thompson S M and Tapia G 2017 Accelerated process optimization for laser-based additive manufacturing by leveraging similar prior studies *IISE Trans.* **49** 31–44

[2] Al-Kharusi G, Dunne N J, Little S and Levingstone T J 2022 The role of machine learning and design of experiments in the advancement of biomaterial and tissue engineering research *Bioengineering* **9** 561

[3] Arduengo J, Hascoët N, Chinesta F and Hascoet J-Y 2024 Open-loop control system for high precision extrusion-based bioprinting through machine learning modeling *J. Mach. Eng.* **24** 103–17

[4] Armstrong A A, Alleyne A G and Johnson A J W 2020 1d and 2D error assessment and correction for extrusion-based bioprinting using process sensing and control strategies *Biofabrication* **12** 045023

[5] Armstrong A A, Norato J A, Alleyne A G and Johnson A J W 2019 Direct process feedback in extrusion-based 3D bioprinting *Biofabrication* **12** 015017

[6] Bai T, Li Y, Shen Y, Zhang X, Zhang W and Cui B 2023 Transfer learning for bayesian optimization: a survey *J. ACM* (arXiv:2302.05927)

[7] Bonatti A F, Chiesa I, Vozzi G and De Maria C 2021 Open-source cad-cam simulator of the extrusion-based bioprinting process *Bioprinting* **24** e00172

[8] Bonatti A F, Vozzi G, Chua C K and De Maria C 2022 A deep learning quality control loop of the extrusion-based bioprinting process *Int. J. Bioprint.* **8** 307–20

[9] Bonatti A F, Vozzi G and De Maria C 2024 Enhancing quality control in bioprinting through machine learning *Biofabrication* **16** 022001

[10] Bonatti A F, Vozzi G, Chua C K and De Maria C 2022 A deep learning approach for error detection and quantification in extrusion-based bioprinting *Mater. Today: Proc.* 70 131–5

[11] Bone J M, Childs C M, Menon A, Póczos B, Feinberg A W, LeDuc P R and Washburn N R Hierarchical machine learning for high-fidelity 3D printed biopolymers *ACS Biomater. Sci. Eng.* **6** 7021–31

[12] Cheng L, Tsung F, Wang Y and Wang A 2017 A statistical transfer learning perspective for modeling shape deviations in additive manufacturing *IEEE Robot. Autom. Letters* **2** 1988–93

[13] Cheng L, Wang K and Tsung F 2020 A hybrid transfer learning framework for in-plane freeform shape accuracy control in additive manufacturing *IISE Trans.* **53** 298–312

[14] Fang J, Wang Z, Liu W, Chen L and Liu X 2024 A new particle-swarm-optimization-assisted deep transfer learning framework with applications to outlier detection in additive manufacturing *Eng. Appl. Artif. Intell.* **131** 107700

[15] Ferreira R C, Sabbaghi A and Huang Q 2019 Automated geometric shape deviation modeling for additive manufacturing systems via bayesian neural networks *IEEE Trans. Autom. Sci. Eng.* **17** 584–98

[16] Francis J, Sabbaghi A, Shankar M R, Ghasri-Khouzani M and Bian L 2020 Efficient distortion prediction of additively manufactured parts using bayesian model transfer between material systems *J. Manufact. Sci. Eng.-Trans. ASME* **142** 051001

[17] Freeman S, Calabro S, Williams R, Jin S and Ye K 2022 Bioink formulation and machine learning-empowered bioprinting optimization *Front. Bioeng. Biotechnol.* **10** 913579

[18] Fu Z, Naghieh S, Xu C, Xu C, Wang C, Sun W, Chen D, Chen D and Chen D 2021 Printability in extrusion bioprinting *Biofabrication* **13** 033001

[19] Ghidini T, Grasso M, Gumpinger J, Makaya A and Colosimo B M 2023 Additive manufacturing in the new space economy: current achievements and future perspectives *Prog. Aerosp. Sci.* **142** 100959

[20] Gillispie G J, Han A, Uzun-Per M, Fisher J, Mikos A G, Niazi M K K, Yoo J J, Lee S J and Atala A 2020 The influence of printing parameters and cell density on bioink printing outcomes *Tissue Eng. A* **26** 1349–58

[21] Gillispie G J, Prim P M, Copus J, Fisher J P, Mikos A G, Yoo J J, Atala A and Lee S J 2020 Assessment methodologies for extrusion-based bioink printability *Biofabrication* **12** 022003

[22] Gironi P, Petraro L, Santoni S, Dedé L and Colosimo B M 2023 A computational model of cell viability and proliferation of extrusion-based 3D-bioprinted constructs during tissue maturation process *Int. J. Bioprint.* **9** 741

[23] Goh G D, Sing S L and Yeong W Y 2021 A review on machine learning in 3D printing: applications, potential and challenges *Artif. Intell. Rev.* **54** 63–94

[24] Groll J *et al* 2018 A definition of bioinks and their distinction from biomaterial inks *Biofabrication* **11** 013001

[25] Gugliandolo S G, Margarita A, Santoni S, Moscatelli D and Colosimo B M 2022 In-situ monitoring of defects in extrusion-based bioprinting processes using visible light imaging *Procedia CIRP* **110** 219–24

[26] Guo T, Holzberg T, Lim C G, Gao F, Gargava A, Trachtenberg J E, Mikos A G and Fisher J P 2017 3D printing plga: a quantitative examination of the effects of polymer composition and printing parameters on print resolution *Biofabrication* **9** 024101

[27] Hoffman A S 2012 Hydrogels for biomedical applications *Adv. Drug. Deliv. Rev.* **64** 18–23

[28] Hölzl K, Lin S, Tytgat L, Van Vlierberghe S, Gu L and Ovsianikov A 2016 Bioink properties before, during and after 3D bioprinting *Biofabrication* **8** 032002

[29] Jin Z, Zhang Z, Shao X and Gu G X 2023 Monitoring anomalies in 3D bioprinting with deep neural networks **9** 3945–52

[30] Knüttel D, Baraldo S, Valente A, Wegener K and Carpanzano E 2022 Transfer learning of neural network based process models in direct metal deposition *Procedia CIRP* **107** 863–8

[31] Krishna D V and Sankar M R 2023 Machine learning-assisted extrusion-based 3d bioprinting for tissue regeneration applications *Ann. 3D Printed Med.* **12** 100132

[32] Kroon D-J 2023 Region growing (available at: www.mathworks.com/matlabcentral/fileexchange/19084-region-growing)

[33] Li J, Wu C, Chu P K and Gelinsky M 2020 3D printing of hydrogels: rational design strategies and emerging biomedical applications *Mater. Sci. Eng. R: Rep.* **140** 100543
[34] Li Z, Zhang Z, Zhang S, Bai Z, Qin R, Huang J, Wang J, Huang K, Zhang Q and Wen G 2023 A novel approach of online monitoring for laser powder bed fusion defects: air-borne acoustic emission and deep transfer learning *J. Manuf. Process.* **102** 579–92
[35] Margarita A, Gugliandolo S G, Santoni S, Moscatelli D and Colosimo B M 2025 A novel solution for real-time in-situ cell distribution monitoring in 3d bioprinting via fluorescence imaging *Biofabrication* **17** 021001
[36] Mehta M and Shao C 2022 Federated learning-based semantic segmentation for pixel-wise defect detection in additive manufacturing *J. Manuf. Syst.* **64** 197–210
[37] Moroni L *et al* 2018 Biofabrication: a guide to technology and terminology *Trends Biotechnol.* **36** 384–402
[38] Nair K, Gandhi M, Khalil S, Yan K, Marcolongo M, Barbee K and Sun W 2009 Characterization of cell viability during bioprinting process *Biotechnol. J.* **4** 1168–77
[39] Ng W L, Chua C K and Shen Y-F 2019 Print me an organ! why we are not there yet *Prog. Polym. Sci.* **97** 101145
[40] Ng W L *et al* 2020 Deep learning for fabrication and maturation of 3d bioprinted tissues and organs *Virtual Phys. Prototyping* **15** 340–58
[41] Ombergen A V *et al* 2023 Bioprinting in microgravity: opportunities, challenges and possible applications in space *Adv. Healthcare Mater.* **12** 2300443
[42] Ouyang L, Yao R, Zhao Y and Sun W 2016 Effect of bioink properties on printability and cell viability for 3D bioplotting of embryonic stem cells *Biofabrication* **8** 035020
[43] Pan S J and Yang Q 2010 A survey on transfer learning *IEEE Trans. Knowl. Data Eng.* **22** 1345–59
[44] Pandita P, Ghosh S, Gupta V K, Meshkov A and Wang L 2021 Application of deep transfer learning and uncertainty quantification for process identification in powder bed fusion *ASCE-ASME J. Risk Uncertain. Eng. Syst. B* **8** 011106
[45] Pandiyan V *et al* 2022 Deep transfer learning of additive manufacturing mechanisms across materials in metal-based laser powder bed fusion process *J. Mater. Process. Technol.* **303** 117531
[46] Pardoe D and Stone P 2010 Boosting for regression transfer *ICML 2010 - Proc., 27th Int. Conf. on Machine Learning* pp 863–70
[47] Pedregosa F *et al* 2011 Scikit-learn: machine learning in python *J. Mach. Learn. Res.* **12** 2825–30
[48] Ramesh S, Deep A, Tamayol A, Kamaraj A, Mahajan C and Madihally S 2024 Advancing 3D bioprinting through machine learning and artificial intelligence *Bioprinting* **38** e00331
[49] Rasmussen C E and Williams C K I 2006 *Gaussian Processes for Machine Learning. Adaptive Computation and Machine Learning* (MIT Press)
[50] Ren J *et al* 2019 Surface variation modeling by fusing multiresolution spatially nonstationary data under a transfer learning framework *J. Manuf. Sci. Eng. Trans. Asme* **141** 011002
[51] Ren J *et al* 2022 Improved modeling of kinematics-induced geometric variations in extrusion-based additive manufacturing through between-printer transfer learning *IEEE Trans. Autom. Sci. Eng.*
[52] Ribeiro A J S, Blokzijl M M, Levato R, Visser C W, Castilho M, Hennink W E, Vermonden T and Malda J 2017 Assessing bioink shape fidelity to aid material development in 3D bioprinting *Biofabrication* **10** 014102
[53] Ruberu K, Senadeera M, Rana S, Gupta S, Chung J, Yue Z, Venkatesh S and Wallace G 2021 Coupling machine learning with 3d bioprinting to fast track optimisation of extrusion printing *Appl. Mater. Today* **22** 100914
[54] Rutz A L, Lewis P L and Shah R N 2017 Toward next-generation bioinks: Tuning material properties pre- and post-printing to optimize cell viability *MRS Bull.* **42** 563–70
[55] Sabbaghi A, Ceccarelli M and Huang Q 2018 Model transfer across additive manufacturing processes via mean effect equivalence of lurking variables *Ann. Appl. Statist.* **12** 2409–2429
[56] Sabbaghi A, Huang Q, Dasgupta T and Dasgupta T 2018 Bayesian model building from small samples of disparate data for capturing in-plane deviation in additive manufacturing *Technometrics* **60** 532–44
[57] Safdar M, Xie J, Ko H, Lu Y, Lamouche G and Zhao Y F 2024 Transferability analysis of data-driven additive manufacturing knowledge: a case study between powder bed fusion and directed energy deposition *J. Comput. Inf. Sci. Eng.* **24** 051010
[58] Santoni S, Gugliandolo S G, Maraldi M, Sponchioni M, Moscatelli D, Moscatelli D and Colosimo B M 2021 3D bioprinting: current status and trends–a guide to the literature and industrial practice *Bio-Des. Manuf.* **5**, 14–42
[59] Schwab A, Levato R, D'Este M, Piluso S, Eglin D and Malda J 2020 Printability and shape fidelity of bioinks in 3D bioprinting *Chem. Rev.* **120** 11028–55
[60] Senanayaka A, Tian W, Falls T and Bian L 2023 Understanding the effects of process conditions on thermal–defect relationship: a transfer machine learning approach *J. Manuf. Sci. Eng.* **145** 071010
[61] Sheng H, Xiao J, Cheng Y, Ni Q and Wang S 2018 Short-term solar power forecasting based on weighted gaussian process regression *IEEE Trans. Ind. Electron.* **65** 300–8
[62] Shilton A, Gupta S, Rana S and Venkatesh S 2017 Regret bounds for transfer learning in Bayesian optimisation *Proc. 20th Int. Conf. on Artificial Intelligence and Statistics, Volume 54 of Proc. of Machine Learning Research (20–22 April)*, ed A Singh and J Zhu (PMLR) pp 307–15
[63] Shin J, Lee Y, Li Z, Hu J, Park S S and Kim K 2022 Optimized 3D bioprinting technology based on machine learning: a review of recent trends and advances *Micromachines* **13** 363
[64] Shin S, Hong S-H, Sainand J and Kim D B 2023 Detecting balling defects using multisource transfer learning in wire arc additive manufacturing *J. Comput. Des. Eng.* **10** 1423–42
[65] Skardal A and Atala A 2014 Biomaterials for integration with 3-D bioprinting *Ann. Biomed. Eng.* **43** 12
[66] Sun J, Yao K, An J, Jing L, Huang K and Huang D 2023 Machine learning and 3D bioprinting *Int. J. Bioprinting* **9** 717
[67] Suntornnond R, Tan E Y S, An J and Chua C K 2016 A mathematical model on the resolution of extrusion bioprinting for the development of new bioinks *Materials* **9** 756
[68] Temirel M, Dabbagh S R and Tasoglu S 2022 Shape fidelity evaluation of alginate-based hydrogels through extrusion-based bioprinting *J. Func. Biomater.* **13** 225
[69] Joy T T, Rana S, Gupta S and Venkatesh S 2019 A flexible transfer learning framework for bayesian optimization with convergence guarantee *Expert Syst. Appl.* **115** 656–72
[70] Trachtenberg J E, Placone J K, Smith B T, Smith T, Piard C M, Santoro M, Scott D W, Fisher J P, Fisher J and Mikos A G 2016 Extrusion-based 3D printing of poly(propylene fumarate) in a full-factorial design *ACS Biomater. Sci. Eng.* **2** 1771–80
[71] Wang J, Aghda N H, Jiang J, Habib A M, Ouyang D and Maniruzzaman M 2022 3D bioprinted microparticles: optimizing loading efficiency using advanced doe technique and machine learning modeling *Int. J. Pharm.* **628** 122302
[72] Weiss K, Khoshgoftaar T and Wang D 2016 A survey of transfer learning *J. Big Data* **3** 05
[73] Yang Q, Zhang Y, Dai W and Pan S J 2020 *Transfer Learning* (Cambridge University Press)
[74] Zanderigo G, Bracco F, Semeraro Q and Colosimo B M 2023 In-situ printability maps (IPM): a new approach for in-situ printability assessment with application to extrusion-based bioprinting *Bioprinting* **36** e00320

[75] Zhang H, Choi J P, Moon S K and Ngo T H 2021 A knowledge transfer framework to support rapid process modeling in aerosol jet printing *Adv. Eng. Inf.* **48** 101264
[76] Zhang W, Deng L, Zhang L and Wu D 2023 A survey on negative transfer *IEEE/CAA J. Automatica Sinica* **10** 305–29
[77] Zhang Z, Zhou X, Fang Y, Xiong Z and Zhang T 2025 AI-driven 3D bioprinting for regenerative medicine: from bench to bedside *Bioactive Mater.* **45** 201–30
[78] Zhu J, Su Z, Wang Q, Hao R, Lan Z, fai Chan F S, Li J and fai Wong S W 2024 Process parameter effects estimation and surface quality prediction for selective laser melting empowered by bayes optimized soft attention mechanism-enhanced transfer learning *Comput. Ind.* **156** 104066
[79] Zhuang F, Qi Z Duan K Xi D Zhu Y Zhu H Xiong H and He Q 2019 A comprehensive survey on transfer learning *CoRR* (arXiv:1911.02685)